

Yu-Hsuan Li

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EDUCATION

National Yang Ming Chiao Tung University

July, 2023 – July, 2027 (expected)

Bachelor of Science in Computer Science GPA: 4.28/4.3 (Rank 2/197)

EXPERIENCE

Software Engineering Intern — Google

June 2026 – Sep 2026

Pixel Core Infrastructure — Test Infrastructure & Observability

- Automated test-configuration generation for an internal regression root-cause isolation service across Android/Pixel kernel builds, cutting triage MTTR from 1–3 days to <2 hours.
- Closed the remediation loop with an event-driven architecture (Cloud Pub/Sub, config propagation polling) to autonomously resume culprit bisection, reducing manual toil from ~1 hour to a 1-click approval.
- Engineered a scheduled SQL data pipeline and operational dashboard that aggregates issue tracker logs into multiple critical metrics on regression volume, bisection accuracy, etc. And surfacing open blocked issues at a glance for engineers to rapidly unblock stuck regressions.

AI4BI Intern, TSMC

Sep, 2025 – June, 2026

- Engineered an automated data pipeline using Python (yfinance) and Cron jobs to maintain and update stock market data in the database, ensuring data availability for downstream analysis.
- Built a data platform with FastAPI and AG Grid, optimizing the processing of large-scale UN Comtrade datasets. Improved data accessibility and interactivity by implementing flexible filtering, aggregation and dynamic visualization.

Research Assistant: Computational Photography Laboratory, NYCU

Sep, 2025 – present

Advised by Prof. Yu-Lun Liu

PUBLICATION

YoCausal: How Far is Video Generation from World Model? A Causality Perspective | 🌐

March, 2026

You-Zhe Xie*, Yu-Hsuan Li*, Jie-Ying Lee, Kaipeng Zhang, Yu-Lun Liu, Zhixiang Wang (* denotes equal contributions)
(Under submission to *NeurIPS 2026*)

Proposed YoCausal, an evaluation protocol for Video Diffusion Models using the Violation of Expectation paradigm.

Developed a two-level benchmark: Level 1 (RSI) for quantifying temporal perception via denoising loss, and Level 2 (CCI) leveraging VLMs to disentangle causal reasoning from temporal bias. Evaluated 13 SOTA VDMs, revealing critical gaps between temporal perception and genuine causal cognition.

PROJECTS

Designing Fun: LLM-based Game Level Generation | 🌐

Jun, 2026

RL final project: trained an LLM to generate engaging levels from gameplay feedback

Python, LLM, RL, SFT

- Decomposed “fun” into three gameplay-grounded rewards such as regret (skill discrimination).
- Two-stage pipeline (SFT → multi-reward GRPO); outperforms vanilla and SFT baselines across all metrics.

Guitar tutor for blind (MC Hackathon Google Team 2nd Place) | 🌐

Oct, 2025

A mobile app designed to teach blind people to learn the guitar from scratch.

FastAPI, Gemini API

- Built the backend web architecture using FastAPI.
- Connected to the Gemini API to interact with users and handle various functionalities.
- Integrated an open-source speech recognition model to distinguish pitch.

Deep Learning Project | 🌐

Aug, 2025

Seven projects from graduate-level courses at NYCU.

- Techniques include: CNN, UNet, ResNet, VAE, Transformer, MaskGIT, Diffusion model.
- Reinforcement learning: Training DQN, PPO, A2C on several tasks.

TECHNICAL SKILLS AND INTERESTS

Languages: C/C++, Python, SQL, JavaScript, HTML, React, C#

Relevant Coursework: Deep Learning, Reinforcement Learning, Machine Learning, Image Processing, Artificial Intelligence, Data Structures & Algorithms, System Administration, Database Systems, Computer Networks

Areas of Interest: Computer Vision, Deep Learning, Machine Learning, Full-Stack Web Development, System Design

Soft Skills: Problem Solving, Self-learning, Adaptability, Collaboration

AWARD

Academic Achievements Award in Fall 2023, 2024, 2025 and Spring 2025 Semesters: Top 5% in the Department

Core Fundamental Course Awards: 2nd/79 in Computer Organization, 1st/90 in Data Structures & Object-oriented Programming, 2nd/195 in Introduction to Operating Systems

University Calculus Award: Top 20 among all 1162 students enrolled in the Calculus Course.

Departmental Capstone Award, June 2026: YoCausal won the Special Excellence Award, ranking 1st/63.